Aasrith Satya Matta

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SUMMARY

AI Engineer with 3+ years of experience designing and integrating AI systems, including machine learning models and AI assisted analytics, into scalable applications. Currently an AI Research Assistant at Auburn University, working on real-time IoT data pipelines, neural network based modeling, and intelligent dashboards for data driven decision making. Experienced in Python, ML workflows, data preprocessing, and model deployment, with growing expertise in Generative AI concepts such as prompt engineering, model fine tuning, and AI driven automation. Strong collaborator in Agile, Git based teams, delivering reliable, production ready AI solutions.

EDUCATION

Auburn University | Master of Science in Computer Science and Software Engineering | Aug 2023- May 2025

Jawaharlal Nehru Technological University | Bachelor of Technology in Electronics and Communication Engineering| Aug 2019- May 2023

WORK EXPERIENCE

AI Research Assistant (TES - Remote) | Sep 2025 - Present | Auburn University, Alabama

- Designed and implemented an interactive IoT-based Sensor Dashboard for the Auburn University College of Poultry Science, integrating BluLog temperature sensors and real time data visualizations to support research on poultry product shelf-life and environmental monitoring.
- Developed and maintained website features and backend services using Node.js, Express.js, JavaScript and SQL-based reporting, improving internal workflow turnaround by 25% through reusable components and cleaner routing.
- Built responsive UI pages with React.js, HTML5, CSS3, Tailwind/Bootstrap (sign-on, dashboards, navigation) and ensured consistent UX across devices.
- Integrated OpenAI large language models (LLMs) into the sensor dashboard to enable context-aware AI interactions, allowing researchers to query shelf-life trends, temperature abuse events, and sensor summaries using natural language.
- Implemented retrieval-augmented generation (RAG)-style context injection, dynamically supplying the LLM with sensor summaries, anomaly flags, and predictive model outputs to generate explainable, data-backed insights.
- Collaborated with faculty and research teams to analyze sensor data, optimize data pipelines, and build AI-assisted insights for predictive shelf life estimation and anomaly detection.
- Tested and validated REST APIs using Postman, improving issue detection speed by 25% and reducing integration defects.
- Used Git for branching and PR-based collaboration troubleshot production issues and improved accessibility (WCAG aligned patterns).

Graduate Teaching Assistant | Aug 2024 - May 2025 | Auburn University, Alabama

- Instructed and mentored 120+ undergraduate students in MATLAB and Python, focusing on programming fundamentals, problem solving, and applied computation.
- Conducted lab sessions, provided one-on-one academic support, and resolved coding/debugging issues to strengthen student understanding.
- Designed, graded, and evaluated assignments, quizzes, and lab projects, streamlining grading workflows and maintaining a 100% on time turnaround.
- Contributed to improved course retention and student feedback scores by enhancing instructional clarity and availability.

Research Assistant(Capstone Project)| Aug 2024 - Dec 2024 | Auburn University college of Pharmacy (AI.Pharm Group), Alabama

- Processed and structured genomic data from 54,000+ cells, converting raw datasets into NumPy based Genomaps for downstream analysis.
- Validated clustering and classification approaches using K-Means simulations and hybrid machine learning pipelines.
- Engineered a ResNet50 + RandomForest classification pipeline in Python, achieving 59.62% accuracy with balanced precision and recall.
- Optimized feature representations extracted from CNN embeddings to improve downstream classification performance and model stability
- Analyzed model outputs and error patterns to identify misclassification trends and potential data limitations.

Development Intern | Sep 2021 - June 2023 | HeadVest, Hyderabad, India

- Developed and maintained a responsive and accessible web interface using React.js, Next.js, HTML, and CSS with Tailwind CSS, ensuring consistent branding across devices for over 200 smart helmet users.
- Developed PHP REST APIs for catalog, authentication, cart operations, addresses, and order placement ensured clean integration between React UI and backend services.
- Improved storefront performance by optimizing React rendering and adding pagination/filtering for product lists, reducing page load time by 20-30%.

- Fixed production issues across checkout and order flows and partnered with stakeholders for iterative releases, reducing checkout related issues by 15-20%.
- Collaborated with designers to translate Figma prototypes and wireframes into production-ready code, applying UI/UX principles that reduced task completion time by 35%.

PROJECTS

Culture Bump:

- Created a dynamic website College Bump Based on a book which helps international students navigate through valuable resources when they land in USA.
- Enhanced user engagement through clean, responsive design and interactive features.

Books and Logistics Database:

- Developed a dynamic web-based platform using PHP and MySQL to simulate a virtual library system, featuring a built in SQL editor for writing and executing custom book related queries.
- Designed an interactive UI with predefined query templates (15+) and contextual descriptions, streamlining user learning and enhancing database query verification for library logistics management.

Fashion Product Classification:

- Built a deep learning based web app using CNNs (TensorFlow/Keras) to classify fashion images by category, type, color, and usage with over 92% accuracy.
- Integrated a Flask backend with a responsive HTML/CSS/JS frontend, enabling real time image uploads and sub 2s prediction latency for enhanced user experience in e-commerce platforms.

TECHNICAL SKILLS

- Core Languages: Python, JavaScript, TypeScript
- AIML & Gen AI: Prompt Engineering, LLM Integration, Neural Networks, Supervised & Unsupervised Learning, Model Evaluation, Performance Optimization
- Frontend: React.JS, Next.JS, HTML, CSS3
- Backend & Databases: PHP, SQL, MongoDB, Node.js, Express.js
- UI/UX & Design Tools: Figma, Adobe Creative Suite and Canva
- Version Control & Testing: Git, GitHub, Jira, Vite
- AI Deployment & DevOps: Docker, CI/CD Pipeline implementation, REST API Integration
- Additional Collaboration Tools: Microsoft Office Suite, Google Workspace and Analytics.

CERTIFICATIONS:

- Foundations of JavaScript, HTML and CSS: Duke University | Coursera
- Fundamentals of MATLAB: MathWorks Inc.
- Machine Learning On-Ramp: MathWorks Inc.
- Google UX Design Specialization: Google | Coursera
- Microsoft Azure AI Essentials: Microsoft | LinkedIn